Eigen-Séta v2 in Rust

Constant-time decapsulation at NIST level 1, 3 and 5 parameter sizes: results report (two rounds)

2026-10-11 · runs: run-20261011T1813Z-bench (19c641d12), run-20261011T1827Z-sidechannel (19c641d12), run-20261011T1955Z-verify (d504af28e) · paper: papers/eigen-seta/

Claims and status

#	Claim	Status	Evidence
1	Correct decapsulation at Toy16, Toy32, L1, L3, L5.	measured: 3,635 decapsulations, no failure	verify + bench
2	pk = ct = 1344 / 1992 / 2632 B; decaps 13–15 / 52–54 / 120–124 s at L1 / L3 / L5.	measured, one host	bench
3	The decaps secret phase executes the same instructions for every key and ciphertext, apart from glibc's allocator.	measured at Toy16 (12 runs) and L1 (4 runs). Evidence, not proof.	verify, sidechannel
4	No timing difference between a fixed ct and fresh ones (FO-rejected included), one key.	measured: $\|t\| \le 2.36$ and ≤ 0.84 in two runs; threshold 4.5	verify, sidechannel
5	Robust to adversarial inputs.	tested: 48 ciphertexts, 14 public keys; no panic or hang	tests
6	NIST categories 1 / 3 / 5.	**conjectured**: binding attack $\approx 2^{\lambda}$ classical, $2^{\lambda/2}$ Grover (shape of AES key search)	paper §6
7	The toy `walk` variant is insecure.	derived from the merged code (key from public degree)	paper §6.2
8	The recovered kernel is always cyclic.	proved (Proposition 3)	paper §4

Defects found and fixed: an LLVM-introduced conditional in safegcd (found by the callgrind probe); a key-generation hang (Tonelli–Shanks accepted z = 0); degenerate secret scalars ≡ −1 (now rejected at keygen); an uncapped basis search on public curves (now capped). The first version of this report gave quantum security as "about λ/2, as for FESTA" without relating it to NIST's definitions, which are themselves AES key search (claim 6).

Correctness, sizes and speed

Set	p bits	pk = ct (B)	keygen (s), n=1	encaps (s)	decaps (s)	correct
Toy16	216	216	0.38	0.005–0.008	0.12–0.19	5/5
Toy32	369	376	2.3	0.009–0.012	0.45–0.59	3/3
L1	1341	1344	32.3	0.24–0.26	13.4–15.0	3/3
L3	1991	1992	118.6	1.03	51.8–53.9	2/2
L5	2630	2632	271.8	2.17–2.20	120.2–124.0	2/2

The verification run added 3,000/3,000 (Toy16), 600/600 (Toy32) and 20/20 (L1) correct decapsulations with fresh keys. Zero failures in 3,000 trials bounds the failure rate only by about 10^{-3}. Negligibility rests on the heuristic O(E/p) argument in the paper.

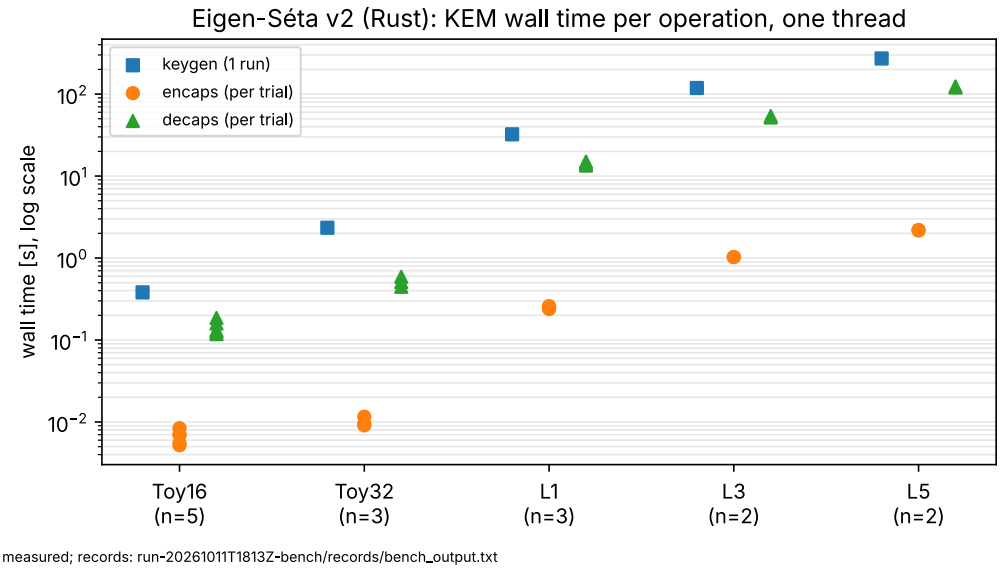


Figure 1: Wall time per operation, all trials, log scale. Measured; run-20261011T1813Z-bench.

Safegcd inversion is 3.9× to 140× faster than Fermat inversion in the same binary. No KEM-level speedup is claimed, because no baseline binary was kept (AGENTS.md §10).

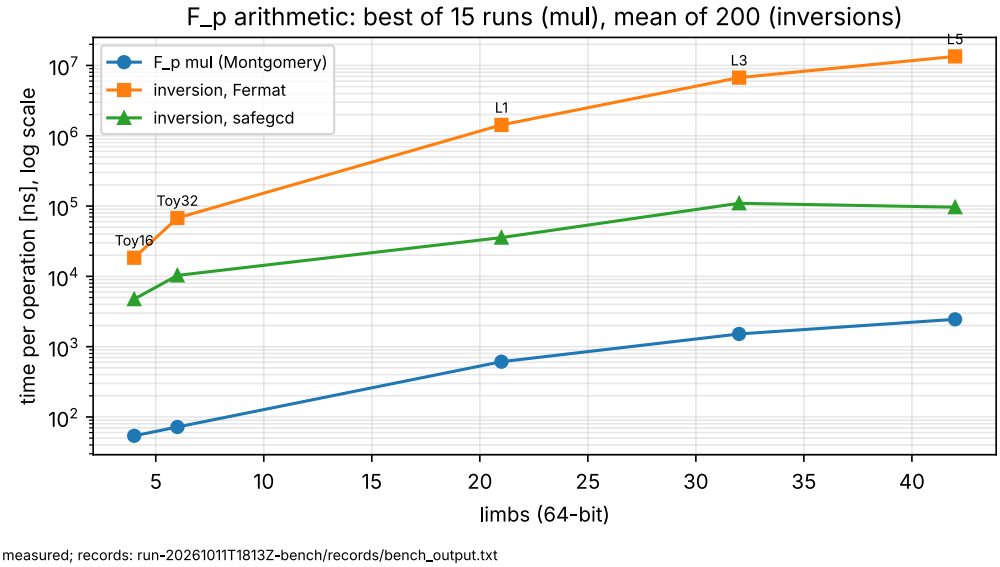


Figure 2: F_p multiplication and inversions against limb count. Measured; run-20261011T1813Z-bench.

Constant-time evidence

Set	Keys	Runs	Non-allocator instructions
Toy16	4	12	1,116,209,216 in every run
L1	2	4	120,996,429,610 in every run

Allocator work, including heap growth through sbrk and brk, varies with the heap state left by earlier variable-time phases. It is equal for the two valid ciphertexts of each key.

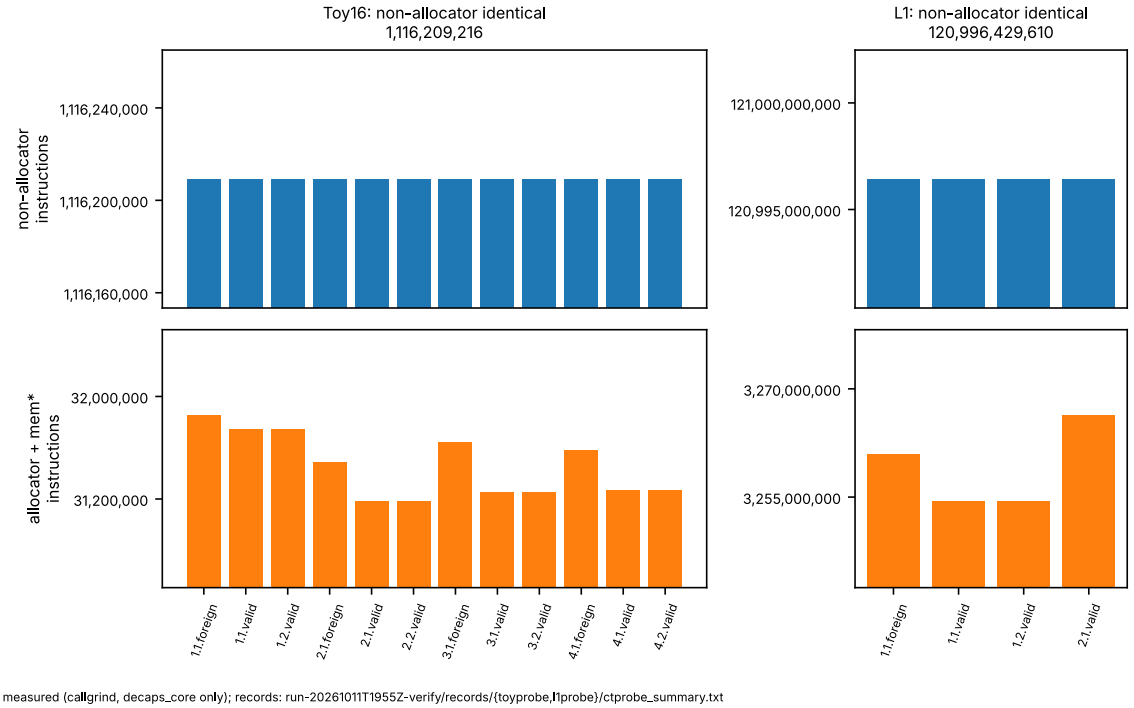


Figure 3: Instruction counts of decaps_core per run. Measured; run-20261011T1955Z-verify.

Limits: instruction counts only, not memory addresses (there are no secret-indexed tables, which is an argument). L3 and L5 were not probed. One compiler and one target. Key generation is variable time by design.

Key-recovery search space

Key generation's public filters accept $2^{-8.13}$, $2^{-8.64}$ and $2^{-9.02}$ of the odd f candidates at L1, L3 and L5. That leaves about 2^{268}, 2^{399} and 2^{525} admissible degrees for an exhaustive search.

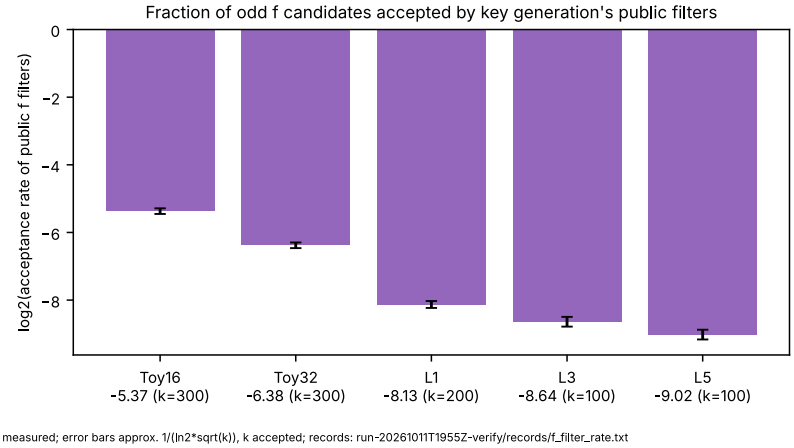


Figure 4: log2 of the filter acceptance rate. Measured; run-20261011T1955Z-verify.

Graphs checked and left unchanged

docs/index-calculus-scoreboard.html, docs/ic/progress-timeline.json, docs/performance-gains/, docs/curves/ and figures/ were checked. None covers isogeny KEMs. docs/curves/ mentions isogenies only as curve-transfer links in the discrete-log registry, and this work adds no registry curve. This report's graphs are new and are drawn from run records by figures/plot.py and figures/plot_verify.py, which only plot recorded values.